# Deep Learning-Based Intelligent Visual Diagnosis of Tomato Yellow Leaf Curl Virus Using Leaf Images: A Review

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**Abstract**

Tomato yellow leaf curl virus (TYLCV) is one of the most damaging pathogens of tomato, and early, accurate field diagnosis is a prerequisite for timely vector control and removal of infected plants. Because its symptoms are visible on leaves, deep-learning analysis of leaf images is an attractive screening route, and the plant-disease imaging literature has grown rapidly. This review synthesises that literature from the perspective of TYLCV. We cover the virus and its diagnostic problem, imaging modalities, the foundations of deep learning, architectures (convolutional networks, transformers, detectors, segmentation and vision-language models), training strategies, datasets, evaluation metrics, calibration, uncertainty and explainability, and deployment on mobile and edge devices. We compare model families and study designs and map the strength of evidence by theme. Three findings recur: near-perfect accuracy on lab-background benchmarks does not transfer to field images; benchmark backgrounds and near-duplicate images can inflate results through shortcuts and leakage; and labels are rarely confirmed by molecular tests. We found no field-validated TYLCV study with molecularly confirmed ground truth. We identify gaps, propose a six-level evaluation maturity ladder and a reporting checklist, and outline research directions for field-ready TYLCV leaf diagnosis.

**Keywords:** tomato yellow leaf curl virus; deep learning; convolutional neural networks; vision transformers; transfer learning; plant disease diagnosis; explainable AI; uncertainty; domain shift; precision agriculture

## 1. Introduction

Tomato (*Solanum lycopersicum*) is one of the most widely grown vegetable crops, and its production is constrained by a wide range of diseases. Among them, tomato yellow leaf curl disease (TYLCD), caused by tomato yellow leaf curl virus (TYLCV), is regarded as one of the most important. When plants are infected early, almost no marketable fruit is produced [1]. The virus is transmitted by whiteflies in a persistent manner [2], can also be seed-transmissible [3], and has spread worldwide through trade and the movement of its vector [4].

No curative treatment exists for an infected plant. Control therefore rests on keeping whiteflies away from the crop, growing resistant cultivars and removing infected plants early [1], [5], [6]. Early removal presupposes early detection. In practice detection relies on trained observers who recognise upward leaf curling, marginal yellowing and stunting, a skill that is scarce and subjective at the scale of commercial production and almost unavailable to many

smallholders. Cameras, in contrast, are cheap and widespread, and a smartphone photograph of a leaf is something almost any grower can produce.

This is the motivation for image-based diagnosis. Deep learning, and in particular convolutional neural networks (CNNs), has transformed image recognition, and since 2016 a large literature has applied it to plant disease. Models trained on leaf photographs reached accuracies above 99% on the PlantVillage benchmark [7], [8], and a large number of surveys now cover the field [9], [10], [11], [12], [13], [14], [15], [16], [17], [18]. At the same time, the earliest large study already reported a sharp drop in accuracy when the model was tested on images taken under different conditions [7], and later work argues that benchmark performance overstates what can be expected in the field [19], [20], [21], [22]. For TYLCV, where management depends on rapid, correct action, an overstated accuracy has real costs: a missed infection allows spread, and a false alarm wastes laboratory tests and may lead to unnecessary removal of healthy plants.

### 1.1 Aim and contributions

This paper is a narrative review that asks a practical question: *how far can deep-learning leaf-image diagnosis be trusted for TYLCV, and what does a trustworthy system require beyond the model itself?* Its contributions are:

- a TYLCV-centred synthesis of the plant-disease imaging literature, spanning foundations, architectures, training, datasets, metrics, calibration, uncertainty, explainability and deployment;
- a comparison of model families, study designs and the strength of evidence by theme, including a per-study summary table;
- an explicit list of research gaps and of the challenges that limit field use; and
- a six-level evaluation maturity ladder, a reporting checklist and a recommended study design for future TYLCV work.

### 1.2 Organisation of the paper

Section 2 describes the review method and positions the paper against existing surveys. Section 3 gives background on TYLCV and the diagnostic problem. Section 4 covers imaging and sensing modalities. Section 5 introduces the deep-learning foundations used in the literature. Section 6 reviews datasets. Section 7 is the main literature review. Section 8 compares the evidence. Sections 9 and 10 present research gaps and challenges. Section 11

proposes directions, a maturity ladder and a recommended study design. Section 12 concludes.

## 2. Review Method and Related Surveys

### 2.1 Search and selection

We searched web and library sources for work on image-based plant disease diagnosis, tomato leaf disease classification, dataset bias and generalisation, detection, segmentation and severity estimation, lightweight deployment, calibration, uncertainty and explainability, and TYLCV biology and management. The scope is roughly 2016–2026, with a small number of foundational deep-learning papers retained where they define methods discussed in the review. Table 1 lists the themes searched and the criteria used to decide what to include.

**Table 1:** Review themes and inclusion criteria

| Item | Description |
|---|---|
| **Themes searched** | Plant disease deep learning; tomato leaf disease; TYLCV biology and management; datasets and bias; detection and severity; lightweight models; calibration and uncertainty; explainability; distribution shift |
| **Time window** | Mainly 2016–2026; earlier foundational methods only where needed |
| **Included** | Studies of leaf-image diagnosis, closely related methods papers, surveys, and TYLCV background needed to define the problem |
| **Excluded** | Work with no image-based diagnosis component; duplicate or superseded reports; sources whose content could not be confirmed |
| **Synthesis** | Narrative, grouped by theme; no meta-analysis because datasets, models and protocols differ |
| **Limits** | Not a systematic review: no registered protocol, search not exhaustive, English-language sources dominate |

Studies were selected for relevance to leaf-image diagnosis and for influence in the field, then grouped by theme and compared qualitatively. Numerical results are quoted only where we are confident of them; where a figure could not be confirmed we describe the finding qualitatively. Because the review is narrative, the absence of a study in our discussion should not be read as proof that none exists; where we state that we did not find a study of a given kind, the statement applies to the sources we consulted.

### 2.2 Relation to existing surveys

Table 2 positions this review against recent surveys. Existing surveys cover plant disease imaging broadly and describe architectures, datasets and open problems; to our knowledge none of those listed focuses on a single virus such as TYLCV, or on the evidence needed

before a screening tool is used in the field. This review therefore adopts a deliberately narrow focus and asks of each body of work what it implies for TYLCV.

**Table 2:** Comparison with existing surveys

| Survey | Scope | Emphasis |
|---|---|---|
| **[9]** | Deep learning in agriculture | Broad agricultural applications |
| **[10], [11], [13]** | Plant disease deep learning | Architectures, datasets and comparisons |
| **[12], [16]** | Image processing and vision-based machine learning | Pipelines and systematic trends |
| **[14], [15], [17]** | Disease and pest detection with deep learning | Detection, CNN reviews and recent advances |
| **[23]** | CNNs for plant disease identification | Field conditions as the main obstacle |
| **[18]** | Diagnosis tools and recommendations | Deployment and tool development |
| **[24], [25]** | IoT, data fusion and remote sensing | Multi-source and large-scale monitoring |
| **This review** | TYLCV-centred, field-evidence focus | Evidence strength, maturity ladder, checklist |

# 3. Tomato Yellow Leaf Curl Virus and the Diagnostic Problem

## 3.1 Virus biology

TYLCV is a monopartite begomovirus in the family *Geminiviridae* [26]. Its virions are twinned (geminate) particles that enclose a circular single-stranded DNA genome of about 2.8 kb, which carries six overlapping open reading frames on the virion-sense and complementary-sense strands, separated by an intergenic region that contains the origin of replication [1]. Table 3 summarises the encoded proteins. The virus replicates by a rolling-circle mechanism that relies on host DNA-replication machinery, and several of its proteins suppress the plant's RNA-silencing defences, which contributes to its ability to establish systemic infection.

**Table 3:** TYLCV open reading frames and main functions

| ORF | Protein | Main function |
|---|---|---|
| **V1** | Coat protein (CP) | Encapsidation, systemic movement, interaction with the vector |
| **V2** | Pre-coat protein | Movement; strong silencing suppressor |
| **C1** | Replication-associated protein (Rep) | Initiates rolling-circle replication |
| **C2** | Transcriptional activator (TrAP) | Transcription control; silencing suppression |
| **C3** | Replication enhancer (REn) | Enhances viral DNA replication |
| **C4** | Pathogenicity factor | Symptom induction, movement, silencing suppression |

### 3.2 Transmission and epidemiology

TYLCV is transmitted by whiteflies of the *Bemisia tabaci* cryptic species complex. Virions are acquired from the phloem during feeding, pass through the stylet and foregut, cross the midgut epithelium into the haemolymph and reach the salivary glands, from which they are inoculated into healthy plants [2]. The efficiency of transmission differs between whitefly species [27], and the virus has also been reported to be seed-transmissible in tomato [3]. This means that the first infected plant in a crop may appear shortly after planting, and that incidence can rise quickly once adult whiteflies colonise the field.

Worldwide spread has been driven by trade in plant material and the global movement of invasive whitefly lineages [4]. Recombination with related begomoviruses adds to the virus's capacity to evolve [28], and related viruses such as tomato leaf curl New Delhi virus show how a different lineage can emerge as a threat to vegetable crops [29]. Taxonomy of the family has itself been revised recently [30].

### 3.3 Symptoms

Infected tomato plants show leaflets that curl upward and inward into a cup shape, yellowing of the leaf margins, shortened internodes, a dwarfed, bushy habit and reduced flowering and fruit set. Severity depends on the age of the plant at infection: plants infected as seedlings are stunted and set almost no fruit, whereas late infection mainly reduces fruit size and quality [1]. Symptoms on the youngest leaves are the most informative, and they develop over days to weeks after inoculation, which means that a photograph taken too early may show a plant that is already infected but not yet symptomatic.

### 3.4 Look-alike conditions

The visual cues of TYLCV are not unique. Several conditions can curl, yellow or distort tomato leaves, and a useful diagnostic system must distinguish them (Table 4). In practice this means that a classifier trained only on TYLCV versus healthy leaves, or against other tomato diseases that look very different from a viral curl, will not tell us how it behaves on the confusing cases that matter in the field.

**Table 4:** Conditions that can resemble TYLCV and how they are normally distinguished

| Condition | Typical visual features | Usual confirmation |
|---|---|---|
| **TYLCV infection** | Upward and inward leaf curling, marginal yellowing, stunting, flower drop | PCR / qPCR [1] |
| **Other begomoviruses** | Similar curling and yellowing; different virus species [29] | PCR and sequencing [4] |

| Condition | Typical visual features | Usual confirmation |
|---|---|---|
| **Nutrient disorders** | Chlorosis (often interveinal or marginal) without systemic curling syndrome | Tissue or soil analysis; response to correction |
| **Herbicide or growth-regulator injury** | Leaf distortion, twisting or cupping, often patchy in the field | Spray history; field pattern |
| **Heat or water stress** | Leaf rolling or scorch, usually reversible | Weather and irrigation context |

### 3.5 Diagnostic methods

Diagnosis can be made at several levels (Table 5). Visual scouting is fast and free but subjective and limited to trained staff. PCR or qPCR on leaf tissue confirms the virus, and sequencing identifies strains and recombinants [4], [28], but both need a laboratory and cannot be applied plant by plant in a large field. Image analysis occupies the space between: it can screen many plants quickly, and its output should be used to prioritise molecular testing rather than replace it.

**Table 5:** Diagnostic methods for TYLCV

| Method | Principle | Strength | Limitation |
|---|---|---|---|
| **Visual scouting** | Trained observer recognises symptoms | Fast, no equipment | Subjective; scarce skill; look-alikes |
| **PCR / qPCR** | Detects viral DNA in tissue | Confirmatory | Laboratory, cost, delay |
| **Sequencing** | Identifies strain and recombinants | Surveillance, evolution | Specialised, slower |
| **Image analysis with deep learning** | Model learns the visual pattern | Scalable screening | Needs data; drops under shift |

### 3.6 Management context

Image screening is one part of integrated management. Resistance genes introgressed from wild tomato relatives are widely used but not fully durable [5]. Whitefly control combines exclusion with nets, reflective mulches, trapping and insecticides, and insecticide resistance in *B. tabaci* is a growing problem [31]. Climate change is expected to alter pest distributions and pressure [32]. Worldwide experience with geminivirus management stresses combining several measures [6]. A screening tool is valuable to the extent that it triggers the right measure at the right time, which is why Section 11 argues for linking image results to vector and risk information.

## 4. Imaging and Sensing Modalities

RGB imaging with consumer cameras is the cheapest and most scalable modality, and nearly all published leaf-image work uses it. Its limitation is that it captures only what is visible:

symptoms must already have appeared, and appearance depends on lighting, camera, background and leaf pose. Hyperspectral and thermal sensors can detect physiological changes, in some cases before visible symptoms, but cost and calibration demands limit them to research and phenotyping settings [33]. Remote sensing from drones and satellites supports monitoring at field and regional scales, although its resolution makes leaf-level diagnosis difficult [25]. Surveys that combine imaging with IoT sensors and environmental data point to multi-modal systems as a longer-term direction [24]. Table 6 summarises the trade-offs.

**Table 6:** Imaging and sensing modalities for leaf disease diagnosis

| Modality | Strengths | Limitations |
|---|---|---|
| **RGB (phone or camera)** | Cheap, ubiquitous, field-ready | Symptom stage only; lighting and background vary |
| **Hyperspectral / thermal** | Pre-symptomatic signals [33] | Cost, calibration, throughput |
| **Remote sensing (drone, satellite)** | Field- to region-scale monitoring [25] | Coarse resolution; symptom ambiguity |
| **Image + IoT / weather** | Context-aware risk and diagnosis [24] | Integration and data-quality burden |

For TYLCV specifically, leaf-level RGB imaging is the practical choice for screening by growers, with remote sensing and multi-modal data most useful for estimating where and when to look. Capture conditions matter. Photographs of the youngest curled leaves against a neutral, consistent background are easiest for a model, but a tool for real growers must work with photographs taken on a plant in the field, with soil, other leaves and uneven light in the frame. This tension between controlled and realistic capture runs through the rest of the review.

## 5. Deep Learning Foundations for Leaf-Image Diagnosis

### 5.1 From pixels to predictions

A digital photograph is a grid of pixel values in three colour channels. A convolutional neural network applies small learned filters across the image to detect local patterns such as edges and textures, then stacks many such layers so that deeper layers respond to larger structures such as lesions, veins or the curled outline of a leaflet. At the end, a classifier layer converts the final features into a score for each class. For a two-class screening problem (TYLCV versus not TYLCV) the network outputs two scores $z_0$ and $z_1$, which are converted to probabilities with the softmax function:

$$p(y = k \mid x) = \exp(z_k) / [\, \exp(z_0) + \exp(z_1) \,] \quad (1)$$

Training adjusts the filter weights to minimise a loss that compares the predicted probabilities with the true labels. The standard loss for classification is the cross-entropy, averaged over N training images with labels $y \in \{0, 1\}$; when classes are imbalanced, class weights $w_y$ can be introduced [34], [35]:

$$L = -(1/N) \Sigma_i w_{y_i} [ y_i \log p(y_i = 1 \mid x_i) + (1 - y_i) \log p(y_i = 0 \mid x_i) ] \quad (2)$$

The weights are updated by gradient descent using back-propagation, usually with an adaptive optimiser; a variant with decoupled weight decay is widely used [36].

### 5.2 Backbone architectures

Early plant-disease studies fine-tuned general-purpose CNNs on leaf photographs [37], [7]. Since then, several architecture families have become standard (Table 7). Inception-style networks use parallel filters of different sizes [38]. Residual networks add shortcut connections that made very deep networks trainable [39], and densely connected networks reuse features across layers [40]. Compound scaling produced the EfficientNet family [41], which has been applied directly to plant leaf classification [42]. For phones and other edge devices, MobileNets and their successors reduce computation [43], [44], [45].

Vision transformers divide an image into patches and process them as a sequence with self-attention [46], and hierarchical variants such as Swin Transformer scale to dense prediction tasks [47]. Transformers are data-hungry, which is a handicap for the small datasets typical of plant disease, but pretrained transformers and hybrid designs are increasingly competitive.

**Table 7:** Representative backbone families used in plant disease studies

| Family | Key idea | Typical use for leaf images |
|---|---|---|
| **Inception [38]** | Parallel multi-scale filters | Transfer learning baselines |
| **ResNet [39]** | Residual shortcut connections | Standard strong baseline |
| **DenseNet [40]** | Dense feature reuse | Fine-tuned classification |
| **EfficientNet [41], [42]** | Compound scaling of depth, width and resolution | Accuracy-efficiency trade-off |
| **MobileNet family [43], [44], [45]** | Depthwise separable convolutions | On-device inference |
| **Vision transformers [46], [47]** | Self-attention over image patches | Large-scale pretraining; emerging in plant studies |

### 5.3 Detection, segmentation and foundation models

Classification assigns one label to a whole image. When several leaves, plants or symptoms appear in one frame, object detectors draw boxes around each: two-stage detectors such as Faster R-CNN [48], single-stage detectors such as SSD and YOLO [49], [50], and

transformer-based detectors [51]. Semantic segmentation assigns a class to every pixel, which allows lesion area and severity to be measured [52]. Promptable foundation models such as the Segment Anything Model delineate objects with minimal supervision [53], and vision-language models align images with text so that classes can be specified in words [54]. These tools are general and are rarely applied to TYLCV specifically; their value for the virus is largely untested.

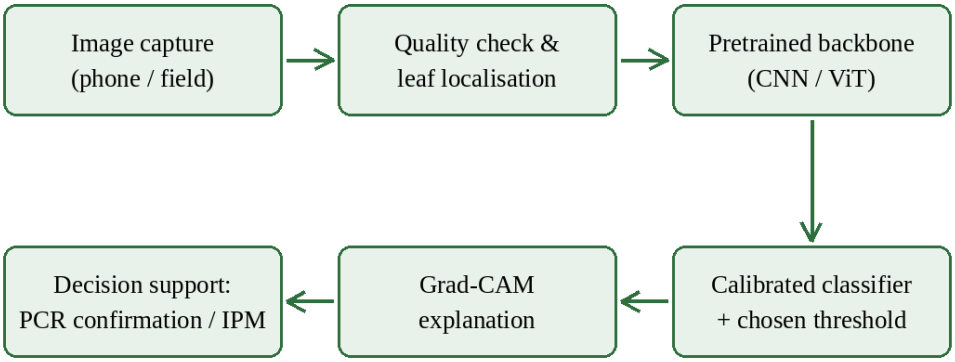



**Figure 1:** Reference pipeline for leaf-image TYLCV screening with calibrated outputs and explanations.

### 5.4 Transfer learning and training strategies

Transfer learning starts from a network pretrained on a large dataset (typically ImageNet) and adapts it to leaf images. It is the default in plant-disease work because datasets are small relative to the models trained on them [55], [19], [56]. Better ImageNet models generally transfer better, although pretraining helps less on some fine-grained tasks [57]. A common schedule trains only the new classification head first and then unfreezes the backbone at a lower learning rate.

Data augmentation enlarges the effective dataset by applying random flips, rotations, crops and brightness or contrast changes [58]; learned augmentation policies and mixing methods are also used [59], [60]. For TYLCV, photometric augmentation needs particular care, because yellowing is the diagnostic cue and aggressive colour changes can erase it. Class imbalance is handled by class-weighted or re-weighted losses [34], [35]. Self-supervised pretraining on unlabelled images, including contrastive methods and self-distillation, offers a route to exploit images without labels [61], [62], [63].

### 5.5 Evaluation metrics

Accuracy alone is a poor summary for screening. When healthy plants outnumber infected ones, a model that always answers “healthy” scores high accuracy while finding no infections. The more informative quantities are the sensitivity (recall), the fraction of infected leaves found, and the specificity, the fraction of healthy or look-alike leaves correctly cleared:

$$\text{Sensitivity} = TP / (TP + FN), \quad \text{Specificity} = TN / (TN + FP) \tag{3}$$

Precision, the fraction of positive calls that are correct, and the F1 score combine these for a given threshold:

$$\text{Precision} = TP / (TP + FP), \quad F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}) \tag{4}$$

The area under the ROC curve summarises ranking quality across all thresholds; for rare positives, precision–recall curves are more informative. Table 8 lists the main metrics and their pitfalls. Protocol matters as much as the metric: random image-level splits can put near-identical photographs of the same plant in training and test sets, whereas external testing at a different site or with a different device measures robustness to distribution shift [64], [65].

**Table 8:** Evaluation metrics and common pitfalls

| Metric | What it tells | Pitfall |
|---|---|---|
| **Accuracy** | Overall fraction correct | Misleading under class imbalance |
| **Sensitivity (recall)** | Share of infected leaves found | Depends on the threshold; hides false alarms |
| **Specificity** | Share of healthy and look-alike leaves cleared | Meaningless without hard negatives |
| **ROC-AUC / PR-AUC** | Ranking quality across thresholds | Not an operating point; PR is better for rare positives |
| **ECE / reliability diagram** | Reliability of predicted probabilities [66] | Needs enough data per bin |
| **External-set performance** | Robustness to shift [64], [65] | Rarely reported |

### 5.6 Calibration, uncertainty and explainability

A diagnostic score is only useful if its probabilities mean what they say. Modern networks tend to be over-confident, and a simple remedy is temperature scaling, which divides the scores by a single parameter T fitted on validation data before the softmax [66]. The quality of calibration is summarised by the expected calibration error, in which predictions are grouped into bins $B_b$ and the gap between accuracy and confidence is averaged:

$$\hat{p} = \text{softmax}(z / T), \qquad ECE = \Sigma_b \, (|B_b| / N) \cdot | \, \text{acc}(B_b) - \text{conf}(B_b) \, | \tag{5}$$

Uncertainty can also be estimated with deep ensembles [67] or Monte-Carlo dropout [68], although predictive uncertainty degrades as the data move away from the training distribution [65]. For explanation, gradient-based saliency such as Grad-CAM highlights the image regions that drove a decision [69] and has been used to check whether tomato disease classifiers attend to lesions [70]; model-agnostic methods such as LIME and SHAP attribute predictions to input features [71], [72]. Explainable pipelines have been used to identify which symptoms drive plant-stress predictions [73], but saliency methods should themselves be validated, because some produce similar maps even for randomised models [74].

## 6. Datasets

PlantVillage is the dominant benchmark: tens of thousands of leaf images across many crop–disease classes photographed in controlled conditions against plain backgrounds [7]. Its convenience produced a large literature but also its main weakness. One analysis showed that PlantVillage backgrounds alone carry enough information to classify images well above chance [22], an instance of the shortcut learning in which networks exploit spurious cues instead of the intended signal [75].

Datasets collected under real conditions are smaller and harder. PlantDoc assembles internet images with annotated bounding boxes for many diseases [76], cassava studies used field images captured on smartphones [77], and tomato greenhouse work used real greenhouse scenes with severity labels [78]. For TYLCV specifically, public collections include a Roboflow Universe dataset [79] and a Kaggle image folder [80]. An inspection of the Kaggle folder for this review found 100 images, all TYLCV-positive, most of them lab-style photographs; it therefore illustrates the data limitations discussed below rather than offering a basis for estimating diagnostic accuracy. Table 9 summarises the main dataset types.

**Table 9:** Dataset types for plant disease image analysis

| Dataset type | Characteristics | Main limitation |
|---|---|---|
| **PlantVillage [7]** | About 54 thousand images, 38 classes, plain lab backgrounds | Background shortcuts [22] |
| **PlantDoc [76]** | About 2.6 thousand in-the-wild images with annotated boxes | Small and uneven classes |
| **Cassava field images [77]** | Smartphone images from real fields | Single crop |
| **Tomato greenhouse images [78]** | Real greenhouse scenes with severity labels | Few diseases; limited public access |
| **TYLCV collections [79], [80]** | Public TYLCV image sets | Composition and labels unverified; negatives often missing |

### 6.1 Properties of an ideal TYLCV dataset

From the limitations above, an ideal TYLCV leaf-image dataset would have the following properties. It would contain images from many sites, cultivars, growth stages, cameras and lighting conditions. It would include healthy leaves and the look-alike conditions of Table 4 as explicit negative classes. A subset would carry molecular confirmation (PCR or qPCR) so that labels are not merely visual opinions. Images of the same plant or scene would be identified so that they can be kept together during splitting. Capture metadata (device, date, location at coarse resolution, crop stage) would be recorded, because it is needed to analyse failures. No public dataset we are aware of meets all of these requirements.

## 7. Literature Review

Figure 2 organises the literature along the axes used in this review: the task, the model family, the data, the measures of trust and the deployment route. The subsections below follow these axes.

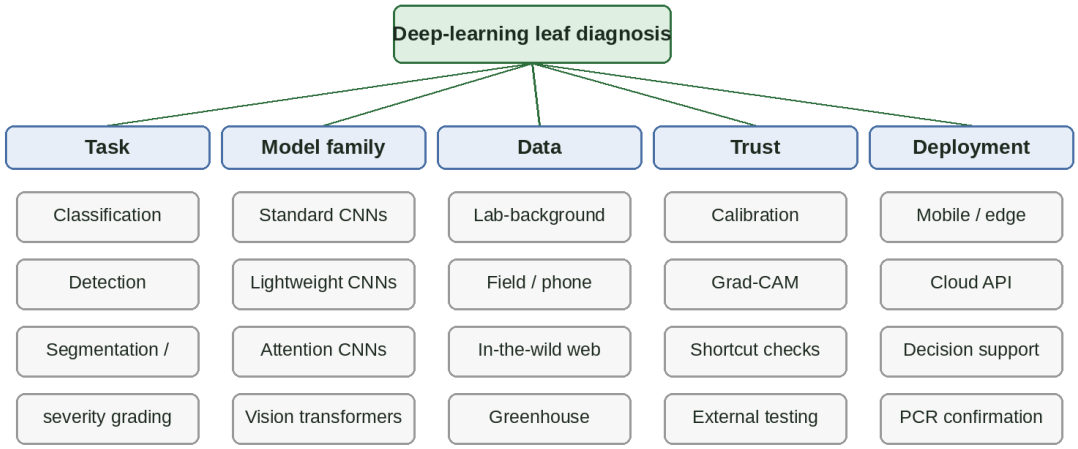



**Figure 2:** Taxonomy of deep-learning leaf-image diagnosis used to organise this review.

### 7.1 Tomato leaf disease classification

Many studies classify tomato leaf diseases, almost all on the PlantVillage tomato classes, with TYLCV as one class among several. Reported work includes comparisons of standard CNNs [81], [70], compact networks for on-device or robotic use [82], lightweight CNNs for modest hardware [83], and attention-augmented networks, including attention-embedded residual networks [84] and squeeze-and-excitation variants [85]. Reported accuracies on this benchmark are uniformly high, which leaves little room to rank methods and little evidence

about field behaviour [55], [8]. We did not find, in the sources consulted, a field-validated study dedicated to TYLCV leaf images with molecular ground truth.

### 7.2 Lessons from other crops

Work on other crops informs what TYLCV studies should expect. Purpose-built nine-layer CNNs reached high accuracy on lab images of many species [86]; lightweight CNNs have been applied to rice diseases and pests [87]; and cassava studies relied on images from real fields [77]. Reviews of CNNs for plant disease conclude that the conditions of real images, rather than the architecture, are the main obstacle [23], and work addressing the current limitations of plant-disease detection stresses dataset size, variety and real-world capture [20]. The common lesson is that models fitted to controlled images do not automatically carry over to the farm.

### 7.3 Generalisation to field conditions

The gap between laboratory and field is the central finding of the literature. A network that reached very high accuracy on held-out PlantVillage images dropped to roughly 31% on images collected under different conditions [7]. Studies of dataset size and variety conclude that capturing diverse conditions matters more than adding more images of the same kind [19], and that classifying individual lesions and spots can be more robust than classifying whole leaves [88]. Work on smartphone capture in the wild shows that training with field-captured images is needed to approach usable field performance [89]. A study re-examining plant-disease characterisation argues for training that reflects practical variation rather than single-source benchmarks [21].

Several mechanisms explain the gap. The first is covariate shift: backgrounds, lighting, camera optics and leaf pose differ between the lab and the field. The second is shortcut learning: when benchmark images share a backdrop that correlates with the class, networks can use it [22], [75]. The third is label shift and selection bias: lab datasets contain clean, typical examples, while field images include occlusion, partial leaves, mixed symptoms and look-alike conditions. Finally, benchmark splits that are random at the image level can place near-identical images in training and test sets, flattering the result.

### 7.4 Detection, segmentation and severity

When several symptoms or plants appear in one frame, detection is more informative than classification. A real-time detector for tomato diseases and pests used region-based detectors with several backbones on field images [90], and a greenhouse system combined detection

with severity estimation [78]. General detectors [48], [49], [50], [51] and segmentation models [52], [53] provide the technical basis for lesion localisation and severity grading. Detection requires bounding-box or mask annotation, which is costly, and no TYLCV-specific annotated set of this kind was identified in the sources consulted.

### 7.5 Lightweight models and deployment

Field use favours models small enough for phones or edge devices [43], [45], [18]. Surveys emphasise practical requirements beyond accuracy: latency, connectivity, user interface and recommendations tied to the diagnosis [18], [24]. Reviews also caution that deployment studies remain scarce relative to benchmark studies [15], [17], [16]. For TYLCV, the practical deployment questions include whether inference runs on the device or in the cloud, how the tool communicates uncertainty, and what the user is told to do after a positive or negative result.

### 7.6 Explainability and trust

Saliency maps were used early to inspect tomato disease classifiers [70] and remain the standard tool for spotting shortcut behaviour: a model that attends to the background rather than the leaf is a warning sign [69], [75]. Saliency is a diagnostic aid, not proof of correctness, and should be combined with out-of-domain testing and, where possible, attribution methods [71], [72], [74]. Trust also requires that explanations are presented in terms a grower can use, for example by highlighting the curled leaf margin rather than an abstract heat-map.

### 7.7 Look-alike discrimination

TYLCV is one of several conditions that curl and yellow tomato leaves, and discriminating among them is the clinically relevant task. Multi-class benchmarks include other tomato diseases, but these are mostly fungal or bacterial spots whose appearance differs strongly from a viral curl, so they are easy negatives [70], [81]. Genuinely hard negatives, such as other begomovirus infections, nutrient disorders and herbicide injury, are rare in public data. Detection of subtle or early symptoms may benefit from lesion-level analysis [88] or from non-visible wavelengths [33], but neither has been demonstrated for TYLCV with molecular ground truth.

### 7.8 Distribution shift and uncertainty

General benchmarks of in-the-wild distribution shift show that models that look excellent in-distribution can fail under shift [64], and that predictive uncertainty degrades as the data move away from the training distribution [65]. These findings match the plant-disease observation

of lab-to-field degradation [7] and argue for external test sets, domain-generalisation methods and uncertainty-aware outputs [67], [68] in TYLCV screening. An uncertainty-aware tool can abstain, or ask for a better photograph, when an image looks unlike its training data.

### 7.9 Label-efficient and foundation-model approaches

Self-supervised methods such as contrastive learning and self-distillation learn visual representations from unlabelled images [61], [62], [63], which could ease the shortage of labelled leaf images, and vision-language models enable classification from text prompts [54]. Promptable segmentation models can assist annotation [53]. We did not find, in the sources consulted, studies applying these approaches specifically to TYLCV leaf images, so their benefit for this virus remains untested.

### 7.10 Data augmentation and synthetic data

Augmentation is the standard remedy for small datasets, and surveys catalogue geometric, photometric and generative approaches [58]. For TYLCV, photometric augmentation needs care because yellowing is the diagnostic cue, and synthetic images cannot replace field images with confirmed labels.

### 7.11 Reported accuracy and what it means

Many studies report accuracies above 95% on tomato leaf classification [8], [55], [84]. Such numbers describe performance on images drawn from the same distribution as the training data, with the same backgrounds and often the same capture sessions. They are informative about model capacity, not about whether a farmer's phone photograph taken in a field will be classified correctly [7], [21]. A fair reading of the literature therefore treats benchmark accuracy as an upper bound rather than an estimate.

## 8. Comparative Analysis of Existing Research

Table 10 summarises key studies discussed in this review, ordered by year of publication. It records the focus, the method and data, and the main point for the present review; it is not a ranking, because the studies used different data and protocols.

**Table 10:** Summary of key studies

| Ref. | Year | Focus | Method and data | Main point for TYLCV |
|---|---|---|---|---|
| **[37]** | 2016 | Leaf-image disease recognition | CNN fine-tuned on web-collected leaf images | Early demonstration of CNN-based leaf diagnosis |
| **[7]** | 2016 | Plant disease classification | AlexNet/GoogLeNet on PlantVillage (54,306 images, 38 classes) | 99.35% on held-out lab images; about 31% on other-condition images |

| Ref. | Year | Focus | Method and data | Main point for TYLCV |
|---|---|---|---|---|
| **[70]** | 2017 | Tomato diseases and visualisation | Pretrained CNNs on tomato classes; saliency maps | Visualisation localises symptoms |
| **[77]** | 2017 | Cassava disease | Transfer learning on field images | Feasibility with real-field images |
| **[90]** | 2017 | Tomato disease and pest detection | Region-based detectors with several backbones; field images | Real-time detection in situ |
| **[82]** | 2017 | Tomato leaf disease | Compact deep networks | Models suited to on-device use |
| **[8]** | 2018 | Disease detection and diagnosis | Several CNNs; open database with 58 classes | 99.53% success on the database |
| **[19]** | 2018 | Dataset size and variety | Transfer learning on varied datasets | Variety of conditions matters most |
| **[81]** | 2018 | Tomato leaf disease | Standard CNNs on tomato classes | Benchmark comparison |
| **[73]** | 2018 | Explainable plant stress | Deep network with explanation | Identifies symptoms driving predictions |
| **[55]** | 2019 | Fine-tuning comparison | Several CNNs fine-tuned on PlantVillage | Similar benchmark accuracy across networks |
| **[89]** | 2019 | Smartphone field images | CNNs on mobile-captured images | Field-captured training data needed |
| **[88]** | 2019 | Lesions and spots | Classification of individual lesions | Can improve robustness |
| **[86]** | 2019 | Purpose-built CNN | Nine-layer CNN on lab images | High lab accuracy |
| **[20]** | 2019 | Limits of current approaches | Analysis of dataset and capture issues | Real-world capture is the obstacle |
| **[76]** | 2020 | In-the-wild dataset | Internet images with annotations | Harder test conditions |
| **[78]** | 2020 | Greenhouse tomato | Detection and severity estimation | Localisation plus severity |
| **[84]** | 2020 | Tomato leaf disease | Attention-embedded residual CNN | Attention helps on benchmark |
| **[83]** | 2020 | Tomato leaf disease | Lightweight CNN | Small model for modest hardware |
| **[21]** | 2020 | Characterising plant disease | Re-examined training strategy | Train for practical variation |
| **[87]** | 2020 | Rice diseases and pests | Lightweight CNN | Compact models viable |
| **[85]** | 2021 | Tomato leaf disease | Residual network with attention module | Reported improved benchmark accuracy |
| **[42]** | 2021 | Plant leaf classification | EfficientNet | Efficient architectures work well |
| **[22]** | 2022 | Dataset bias | Background-only experiment | PlantVillage backgrounds carry class information |

### 8.1 Model families

Table 11 compares the main model families. The families differ in capacity, data requirements and suitability for devices, but for TYLCV the decisive differences are not architectural: most of these model families have been evaluated only on benchmark data.

**Table 11:** Comparison of model families

| Model family | Strengths | Limitations for TYLCV |
|---|---|---|
| **Standard CNNs [7], [39], [38], [40]** | Strong, well understood, easy to fine-tune | Large; benchmark-tuned |
| **Lightweight CNNs [41], [43], [45]** | Fast and small; mobile-friendly | Capacity trade-offs; little field evidence |
| **Attention CNNs [84], [85]** | Small gains on benchmarks | Gains within benchmark saturation |

| Model family | Strengths | Limitations for TYLCV |
|---|---|---|
| **Vision transformers [46], [47]** | Strong general vision results | Data-hungry; little plant field evidence |
| **Detectors [48], [49], [50], [51]** | Localise several symptoms per image | Costly box annotation; few TYLCV sets |
| **Segmentation and foundation models [52], [53], [54]** | Lesion delineation; promptable, zero-shot | Untested for TYLCV |

### 8.2 Study designs

Table 12 compares study designs by what they can and cannot establish. The key distinction is not architecture but evaluation: lab-background benchmarks reward capacity and tuning, whereas field-style, duplicate-aware and externally validated evaluations measure what matters for TYLCV screening.

**Table 12:** Comparison of study designs

| Study design | Typical evidence | Main limitation |
|---|---|---|
| **Lab-background classification [7], [8], [55]** | Very high benchmark accuracy | Background shortcuts; large drop off-distribution [7], [22] |
| **Attention and lightweight CNNs [84], [85], [83]** | Small gains over standard CNNs | Gains within benchmark saturation |
| **Detection and severity [90], [78]** | Localises symptoms in real scenes | Costly annotation; few TYLCV sets |
| **Field and smartphone capture [89], [77], [19]** | Most realistic accuracy | Smaller datasets; lower accuracy |
| **Explainability checks [69], [70]** | Exposes shortcuts | Qualitative; not proof |

Architectures themselves matter less than data. Comparative fine-tuning studies find that several mainstream networks reach similar benchmark accuracy [55], while studies of data variety find large effects from the diversity of training images [19]. For TYLCV, the available evidence is therefore dominated by benchmark results that do not isolate the virus from look-alike conditions. A second observation concerns reporting: many papers give a single accuracy on a random split, without confidence intervals, external validation, specificity against look-alikes or calibration. Without these, two methods that differ by one percentage point on a saturated benchmark cannot be ranked, and neither can be trusted in the field [66], [17]. Reviews consistently call for standardised protocols and more realistic datasets [15], [14], [18], [23].

### 8.3 Strength of evidence by theme

Table 13 summarises how strong the evidence is for each theme, separating what is well established from what remains open.

**Table 13:** Evidence map by theme

| Theme | Strength of evidence | Key sources |
|---|---|---|
| **Benchmark accuracy** | Strong: many studies, consistent results | [7], [8], [55] |
| **Field generalisation** | Moderate: mostly other crops | [19], [89], [77], [21] |
| **Dataset bias and shortcuts** | Moderate: few dedicated studies | [22], [75] |
| **Detection and severity** | Limited for tomato | [90], [78] |
| **Calibration and uncertainty** | Strong in general ML; rare in plant studies | [66], [65], [67] |
| **Foundation and self-supervised models** | None found for TYLCV | [54], [63], [61] |
| **TYLCV-specific field validation** | None found in the sources consulted | — |

## 9. Research Gaps

Eight gaps follow from the review. Each is stated with the evidence behind it.

**Gap 1: Molecularly confirmed labels.** Public image sets are commonly labelled by visual inspection. Large-scale TYLCV image sets with PCR-confirmed ground truth are, to our knowledge, not publicly available [1], [80]. Without them, a model may learn the labeller's opinion rather than the infection state.

**Gap 2: Negatives and look-alikes.** Evaluation needs healthy leaves and conditions that mimic TYLCV (other begomoviruses, nutrient and herbicide injury); otherwise specificity is unknown [1], [33].

**Gap 3: Field generalisation.** Accuracy collapses when conditions change [7], [19], and backgrounds leak labels [22], [75].

**Gap 4: Leakage-aware splitting.** Near-duplicate and same-plant images inflate scores unless splits are group-aware; most reviewed studies do not report such precautions.

**Gap 5: Calibration, uncertainty and thresholds.** Most studies report accuracy only, not calibrated probabilities, uncertainty or operating points [66], [67].

**Gap 6: Early and pre-symptomatic detection.** RGB models detect visible symptoms; earlier detection needs richer sensing [33], [24], [25].

**Gap 7: Foundation and label-efficient models.** Self-supervised and vision-language approaches are untested for TYLCV [54], [63].

**Gap 8: Deployment evidence.** Few studies evaluate end-to-end field use including user decisions [18], [15].

## 10. Challenges and Limitations

Class imbalance and small field datasets limit training. Symptom expression varies with cultivar, plant age, temperature and co-infections, so appearance is not a stable label [1]. Imaging conditions (illumination, camera, leaf pose) differ between users, and even the same grower may photograph leaves differently from day to day. Saliency maps can be misleading if over-interpreted [69], [74]. Labels derived from visual inspection carry noise, and molecular confirmation at scale is costly. Regulatory and trust issues arise if a screening tool influences decisions about removing plants.

Ethical and practical considerations also apply. Geo-tagged farm images can reveal field locations and should be handled with consent and appropriate privacy; access to phones, connectivity and extension support differs between growers, so tools should degrade gracefully and be tested with smallholders as well as commercial farms [18]. Automated screening should be presented as decision support with its uncertainty, not as a replacement for expert or laboratory diagnosis, and its limits should be stated to users.

**Threats to validity in the published literature.** Several recurring problems limit what published results can show. *Selection bias:* benchmark images are typical, well-framed examples, whereas field images include partial leaves, mixed symptoms and clutter. *Label noise:* labels assigned by visual inspection can be wrong, particularly for early or mild symptoms and for look-alike conditions, and a model trained on them inherits the errors. *Data leakage:* near-duplicate images, images of the same plant at different times, or images from the same capture session can fall on both sides of a split. *Tuning on the test set:* when hyperparameters or checkpoints are chosen by test performance, the reported number is optimistic. *Class balance and prevalence:* sensitivity and specificity do not depend on prevalence, but predictive values do, and the prevalence of TYLCV in a field is rarely the balanced proportion of a benchmark; a model that looks precise on a balanced test set can produce many false alarms where most plants are healthy (Section 11.3). *Reporting:* single training runs without seeds, intervals or code limit reproducibility.

Finally, this review is limited by its narrative design, its reliance on published summaries, and the absence of TYLCV-specific field studies to compare. We have tried to be explicit about where claims rest on evidence from other crops or from general machine-learning studies.

# 11. Future Research Directions

## 11.1 Evaluation maturity ladder

We propose a six-level evaluation maturity ladder (Figure 3). Most published studies sit at the first level. Field use of a TYLCV screening tool should rest on evidence from at least levels L2–L4, with L5 and L6 needed before the tool influences management decisions.

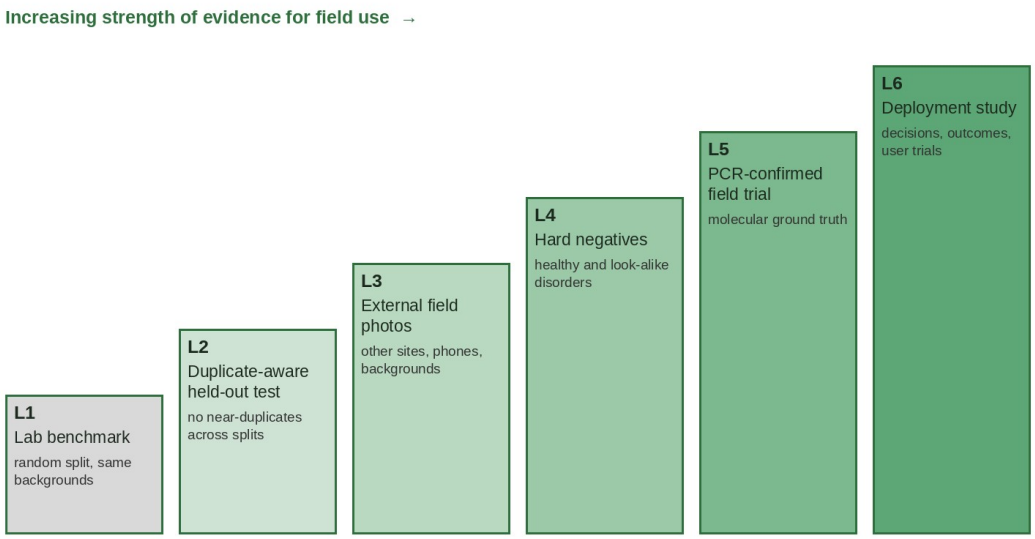



**Figure 3:** Evaluation maturity ladder for TYLCV leaf-image diagnosis, from lab benchmark to deployment study.

## 11.2 From gaps to directions

Table 14 maps the research gaps to concrete directions.

**Table 14:** Research gaps and proposed directions

| Gap | Direction |
|---|---|
| **PCR-confirmed labels (1)** | Multi-site, multi-device dataset with molecular ground truth |
| **Negatives and look-alikes (2)** | Include healthy leaves and hard-negative disorders; report per-condition false positives |
| **Field generalisation (3)** | External field sets; background-only and Grad-CAM shortcut checks [69], [22] |
| **Leakage (4)** | Group-aware splits; document duplicate handling |
| **Calibration, uncertainty (5)** | Temperature scaling and ensembles; thresholds fixed on validation data [66], [67] |
| **Early detection (6)** | Hyperspectral or multi-modal studies [33], [24] |
| **Foundation models (7)** | Self-supervised pretraining and vision-language tests [61], [63], [54] |
| **Deployment (8)** | On-device models and field user studies [45], [18] |

### 11.3 Predictive value and the cost of errors

The usefulness of a positive result depends on how common infection is. The positive predictive value (PPV), the probability that a flagged plant is truly infected, follows from Bayes' rule in terms of the sensitivity, the specificity and the prevalence π of infection:

$$\mathrm{PPV} = (\mathrm{Sensitivity} \cdot \pi) \,/\, [\, \mathrm{Sensitivity} \cdot \pi + (1 - \mathrm{Specificity}) \cdot (1 - \pi) \,] \qquad (6)$$

As a purely illustrative example (hypothetical numbers, not results from any study), consider a screening model with 95% sensitivity and 95% specificity. If 5% of plants are infected, the PPV is $0.95 \times 0.05 \,/\, (0.95 \times 0.05 + 0.05 \times 0.95) = 50\%$: half of the flagged plants are false alarms. If 20% of plants are infected, the PPV rises to about 83%. The same model therefore behaves very differently early in an outbreak, when few plants are infected, than late in it. This is why a screening result should be interpreted together with local prevalence and vector pressure, why the decision threshold should reflect the relative costs of missed infections and false alarms, and why reporting sensitivity and specificity separately, with confidence intervals, is more informative than reporting accuracy.

### 11.4 Combining images with context

The limitations of image-only diagnosis suggest a complementary design: combine a calibrated image classifier with contextual evidence such as whitefly density, weather and local virus prevalence, which together determine the prior probability of infection [24], [1]. In such a system the image model need not be perfect. A positive screen raises the priority of PCR testing and of vector scouting, and a negative screen in a high-risk context should not suppress monitoring. This framing also clarifies the cost of errors: a false negative delays removal of infected plants and allows spread, whereas a false positive costs a laboratory test. The decision threshold should therefore be chosen with these costs in mind rather than by default.

### 11.5 Reporting checklist

Table 15 gives a reporting checklist consistent with the ladder, so that readers can judge which level a study has reached and what it leaves out.

**Table 15:** Proposed reporting checklist

| Item | Requirement |
|---|---|
| **Labels** | PCR or qPCR-confirmed where possible |
| **Negatives** | Healthy leaves plus look-alike disorders |
| **Splits** | Group-aware; no near-duplicates across splits |
| **External test** | Different sites, cameras and backgrounds |

| Item | Requirement |
|---|---|
| **Metrics** | Sensitivity, specificity and AUC with confidence intervals; calibration |
| **Shortcut checks** | Grad-CAM and a background-only baseline |
| **Reporting** | Threshold fixed on validation data; code and data availability |

### 11.6 Recommended design for a new TYLCV study

Putting the above together, we recommend the following sequence. (1) Define the task and the negative classes, including look-alike disorders. (2) Collect images across sites, devices and growth stages, with a PCR-confirmed subset. (3) Remove duplicates and assign near-duplicate groups before splitting. (4) Choose a lightweight pretrained backbone [45], [41] and at least one larger baseline [39], [46]. (5) Train with staged fine-tuning, class weighting and small photometric augmentation [55], [34], [58]. (6) Fit temperature scaling and fix the decision threshold on validation data only [66]. (7) Evaluate on the internal test set, an external field set and hard negatives, reporting confidence intervals. (8) Run background-only and Grad-CAM shortcut checks [69], [22]. (9) Report against Table 15 and state the level reached on the ladder.

### 11.7 Roadmap

In the short term, the most valuable work is data: a multi-site TYLCV image collection with healthy and look-alike negatives, and a PCR-confirmed subset. In the medium term, standard baselines and the evaluation protocol above can be applied to that data, including external field tests, calibration and shortcut checks. In the longer term, label-efficient and foundation-model approaches, multi-modal systems that combine images with vector and weather data, and field deployment studies can be explored.

### 11.8 Use scenarios

Requirements differ by who uses the tool and why (Table 16). A smallholder needs a model that runs on a phone with poor connectivity and gives advice in plain language. A nursery or greenhouse needs high throughput and few false alarms, because each flagged transplant triggers a test or a loss. Extension services need aggregate, geo-referenced data with appropriate privacy. For seedlings and transplants, image screening cannot find plants that are infected but not yet symptomatic, and seed-borne infection has been reported [3], so image screening must be paired with sampling and molecular testing.

**Table 16:** Use scenarios and design implications for a TYLCV screening tool

| Scenario | Typical user | Priorities | Design implications |
|---|---|---|---|
| **Smallholder with a phone** | Grower | Low cost, low connectivity, clear advice | On-device lightweight model [45]; calibrated output; simple language |
| **Commercial greenhouse or nursery** | Technician | Throughput; few false alarms; record keeping | Consistent capture station; threshold tuned for high specificity; link to scouting records |
| **Extension and surveillance** | Agency or researcher | Aggregated, geo-referenced reports | Metadata, privacy controls, uncertainty reporting; molecular confirmation of a subset |
| **Seedling and transplant production** | Producer | No infected plants sold | Sampling and PCR alongside imaging [3]; image screening for symptomatic plants only |

## 12. Conclusion

Deep learning can recognise TYLCV-like symptoms in leaf images, but the evidence that it works in the field is weak. High benchmark accuracies are not reliable guides to field behaviour because of lab backgrounds, label shortcuts, duplicate leakage and the absence of molecularly confirmed negatives, and we found no field-validated TYLCV study with PCR ground truth. Progress requires better data and stricter evaluation more than new architectures. With group-aware splits, external field tests, calibrated thresholds, uncertainty estimates and explanation checks, image-based screening can become a trustworthy first step ahead of PCR confirmation and vector management. The maturity ladder and checklist offered here are intended to make that progress measurable.

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## Declarations

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## Abbreviations

| | |
|---|---|
| **AUC** | Area under the ROC curve |
| **CNN** | Convolutional neural network |

| | |
|---|---|
| **ECE** | Expected calibration error |
| **FN / FP** | False negative / false positive |
| **Grad-CAM** | Gradient-weighted class activation mapping |
| **IoT** | Internet of Things |
| **PCR** | Polymerase chain reaction |
| **PPV** | Positive predictive value |
| **PR** | Precision–recall |
| **qPCR** | Quantitative PCR |
| **RGB** | Red–green–blue colour imaging |
| **ROC** | Receiver operating characteristic |
| **TN / TP** | True negative / true positive |
| **TYLCD** | Tomato yellow leaf curl disease |
| **TYLCV** | Tomato yellow leaf curl virus |
| **ViT** | Vision transformer |
| **YOLO** | You only look once (object detector) |

## Appendix A: Glossary of Terms

**Table 17:** Glossary

| Term | Meaning |
|---|---|
| **Backbone** | The feature-extracting part of a network (for example ResNet or MobileNet) to which a task-specific head is attached |
| **Calibration** | Agreement between predicted probabilities and observed frequencies [66] |
| **Domain shift** | A change between the data a model was trained on and the data it is used on, such as lab to field [64] |
| **ECE** | Expected calibration error: average gap between confidence and accuracy over probability bins |
| **Fine-tuning** | Continuing to train a pretrained network on a new dataset |
| **Grad-CAM** | A method that highlights image regions that contributed most to a class score [69] |
| **Group-aware split** | A train/validation/test split that keeps near-duplicate or same-plant images together |
| **Hard negative** | A non-target example that resembles the target, such as a look-alike disorder |
| **Overfitting** | Fitting the training data so closely that performance on new data falls |
| **PPV** | Positive predictive value: the fraction of flagged cases that are truly positive |
| **Self-supervised learning** | Learning representations from unlabelled data [61] |
| **Sensitivity / specificity** | Fraction of true positives found / fraction of true negatives cleared |
| **Shortcut learning** | Using spurious cues, such as backgrounds, instead of the intended signal [75] |
| **Transfer learning** | Reusing a model pretrained on one task as the starting point for another [56] |

## Appendix B: Questions to Ask of a TYLCV Image-Diagnosis Study

Readers and reviewers can use the following questions to judge which level of the maturity ladder a study has reached.

- Were the labels confirmed by PCR or qPCR, or assigned by visual inspection?
- Did the negative class include healthy leaves and look-alike conditions, or only easy negatives?
- Were near-duplicate and same-plant images kept on one side of the split?
- Was there an external test set from different sites, cameras and backgrounds?
- Were sensitivity and specificity reported with confidence intervals, not only accuracy?
- Were probabilities calibrated, and was the threshold fixed without using the test set?
- Were shortcut checks run, for example Grad-CAM and a background-only baseline?
- Were code, data and licences available, and were results reported over several random seeds?

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